

1. START

class person

DATA:

name

age

METHOD getpersonDetails()

INPUT name

INPUT age

END class

class student EXTENDS person

DATA:

registerNumber

marks[5]

METHOD getstudentDetail()

INPUT registerNumber

for i=0 to 4

INPUT marks[i];

END FOR

END METHOD

METHOD calculateAverage()

sum ← 0

FOR i=0 to 4

sum ← sum + marks[i]

END FOR

average ← sum/5

RETURN average

END METHOD

END CLASS

class ResearchStudent EXTENDS student

DATA:

researchArea

METHOD getResearchDetails()

INPUT researchArea

END METHOD

METHOD display()

PRINT "Name:", name

PRINT "Age:", age

PRINT "Register Number:", reg-no

PRINT "Research Area:", researchArea

PRINT "Average Marks:", calculateAverage()

END METHOD

END CLASS

MAIN

CREATE object rs OF ResearchStudent

CALL rs.getpersonDetails()

CALL rs.getstudentDetails()

CALL rs.getResearchDetails()

CALL rs.calculateAverage()

CALL rs.display()

END

2) Missing Logic

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$

DELETE $A[j]$

END IF

END FOR

END FOR

PRINT Modified Array

END

Missing Logic:

1. $i+1$

2. DELETE $A[j]$ (or REMOVE $A[j]$)

3. Pseudo code:

CLASS Area

METHOD area (length)

 RETURN length * length

END METHOD

METHOD area (length, breadth)

 RETURN length * breadth

END METHOD

END CLASS

Incorrect line: RETURN length * breadth (inside METHOD area(length))

correct line: RETURN length * length

4. Pseudo code:

BEGIN

sum $\leftarrow$ 0

FOR $i \leftarrow 1$ to 5

IF $i \bmod 2 == 0$

sum $\leftarrow$ sum + i

ELSE

sum $\leftarrow$ sum - i

END IF

END FOR

PRINT sum

END

Expected output:

i	condition	operation	sum
1	odd	$0 - 1$	-1
2	Even	$-1 + 2$	1
3	odd	$1 - 3$	-2
4	Even	$-2 + 4$	2
5	odd	$2 - 5$	-3

The final output is -3.

5. pseudo code;

BEGIN

READ N

READ ARRAY A

$largest \leftarrow -\infty$

$secondlargest \leftarrow -\infty$

FOR $i \leftarrow 0$ to $N-1$

IF $A[i] > largest$ THEN

$secondlargest \leftarrow largest$

$largest \leftarrow A[i]$

ELSE IF $A[i] > secondlargest$ AND $A[i] \neq largest$ THEN

$secondlargest \leftarrow A[i]$

END IF

END FOR

IF $secondlargest = -\infty$ THEN

PRINT "second largest distinct element not found"

ELSE

PRINT $secondlargest$

END IF

END

6. Pseudo code:

BEGIN

INTERFACE Payment

METHOD pay(amount)

END INTERFACE

CLASS CreditCard IMPLEMENTS Payment

METHOD pay(amount)

PRINT "Payment made using credit card:", amount

END METHOD

END CLASS

CLASS UPI IMPLEMENTS Payment

METHOD pay(amount)

PRINT "Payment made using net Banking:", amount

END METHOD

END CLASS

MAIN

READ amount

READ choice

if choice = 1 THEN

Payment ← CreditCard

ELSE IF choice = 2 THEN

Payment ← UPI

Else

Payment ← NetBanking

CALL Payment.pay(amount)

END

7 Complete Pseudo code:

```
CLASS Animal
```

```
    METHOD speak()
```

```
        PRINT "Animal"
```

```
    END METHOD
```

```
END CLASS
```

```
CLASS Dog inherits Animal
```

```
    METHOD speak()
```

```
        PRINT "Barks"
```

```
    END METHOD
```

```
END CLASS
```

```
Animal obj ← new Dog()
```

```
obj.speak()
```

Blank 1 : inherits (extends)

Blank 2 : new Dog() (runtime polymorphism)

8. Correct Pseudo code:

CLASS Employee

DATA

id

name

CONSTRUCTOR()

this.id ← id

this.name ← name

END CONSTRUCTOR

END CLASS

9. Complete Pseudo code:

BEGIN

CLASS Vehicle

DATA

vehicleNo

ownerName

METHOD display()

PRINT vehicleNo

PRINT ownerName

END METHOD

END CLASS

CLASS car inherits Vehicle

DATA

fuelType

METHOD display()

PRINT vehicleNo

PRINT ownerName

PRINT fuelType

END METHOD

END CLASS

CLASS Truck inherit vehicle

DATA

loadCapacity

Method display()

PRINT vehicleNo

PRINT ownerName

PRINT loadCapacity

END METHOD

END CLASS

MAIN

READ N

DECLARE vehicle list[N]

FOR $i \leftarrow 0$ to $N-1$

READ type

IF type = "car" THEN

obj $\leftarrow$ new car

READ vehicleNo, ownerName, fuelType

ELSE

obj $\leftarrow$ new Truck

READ vehicleNo, ownerName, loadCapacity

ENDIF

list[i] $\leftarrow$ obj

END FOR

FOR $i \leftarrow 0$ TO $N-1$

CALL list[i].display()

ENDFOR

END